Example 1.7 The Kepler problem describes the motion of a body in the plane moving under gravitational force exerted by a second, fixed body (located at the origin); it has the energy $E(x, y, \dot{x}, \dot{y}) = \dot{x}^2/2 + \dot{y}^2/2 - \frac{1}{\sqrt{x^2+y^2}}$. Note that a consequence of fixing one of the bodies in the Kepler problem is that the two components of the total momentum vector, i.e. $(m\dot{x}, m\dot{y})$, are obviously no longer conserved; however, due to the fact that the potential energy is rotationally invariant (dependent only on the distance of the moving particle from the origin), the *angular momentum* of the system is conserved. The two conserved quantities, energy and angular momentum, mean that the Kepler problem is an integrable system.

In polar coordinates $(x, y) = (r\cos\theta, r\sin\theta)$, the Lagrangian L for the Kepler problem is

$$\begin{aligned} L &= K - U \\ &= \frac{1}{2}(\dot{r}\cos\theta - r\dot{\theta}\sin\theta)^2 + \frac{1}{2}(\dot{r}\sin\theta + r\dot{\theta}\cos\theta)^2 + \frac{1}{r} \\ &= \frac{\dot{r}^2}{2} + \frac{r^2\dot{\theta}^2}{2} + \frac{1}{r}. \end{aligned}$$

The form of Hamilton's equations is the same no matter what variables we work in (as long as the change of variables is regular, i.e., it can be inverted). Thus, in polar coordinates,

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{r}} = \frac{\partial L}{\partial r}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{\theta}} = \frac{\partial L}{\partial \theta}.$$

Working these out directly, one gets

$$\begin{aligned} \ddot{r} &= -\frac{1}{r^2} + r\dot{\theta}^2, \\ 0 &= \frac{\mathrm{d}}{\mathrm{d}t}(r^2\dot{\theta}). \end{aligned}$$

In terms of the original variables, $l_z = x\dot{y} - y\dot{x}$ is the coordinate of the angular momentum which points out of the plane of motion. Expressed in polar coordinates this is

$$\begin{aligned} l_z &= (r\cos\theta)(\dot{r}\sin\theta + r\dot{\theta}\cos\theta) - (r\sin\theta)(\dot{r}\cos\theta - r\dot{\theta}\sin\theta) \\ &= r^2\dot{\theta}(\cos^2\theta + \sin^2\theta) = r^2\dot{\theta}, \end{aligned}$$

so the second equation of motion in polar coordinates just expresses the constancy of angular momentum. Taking this quantity as fixed, we may write the remaining equation as $\ddot{r} = -\frac{1}{r^2} + \frac{l_z^2}{r^3}$. The equation for r is that of a one degree-of-freedom